Agentic File System Organizer

Complementing my literature review on “Building Effective AI Agents”, I am building an agentic application that solves a problem I personally face and assume many others share. After a work session I find my desktop and download folders filled with scattered files, unorganized with weird names. The AI agent will have access to a specified folder to start reorganizing and creating relevant folders. The agent will be able to read a variety of file types ranging from txt, pdf, or images to understand their context and store them accordingly. Claude by anthropic have just release a similar idea “Claude Cowork”, where Claude can interact with your desktop and read through the files.

So what makes my Agent Organizer novel? I am building it completely using local models. I do not want to share personal files on my desktop with Anthropic or OpenAI, so I am using smaller models that can run on a new MacBook machine with newer chips. The goals is to create a free to use fully local agentic application that can run in your command line. The features of the applications are still not ironed out.

To evaluate the effectiveness of the Agentic File System Organizer, I will benchmark the system across three dimensions: organization accuracy, efficiency, and usability. The goal is to compare the agent’s performance against baseline alternatives and measure whether a local agentic approach can reliably organize unstructured personal files. I will also run user studies, on participants machines and get their review.

A flowchart illustrating the agent's task planning and execution process.

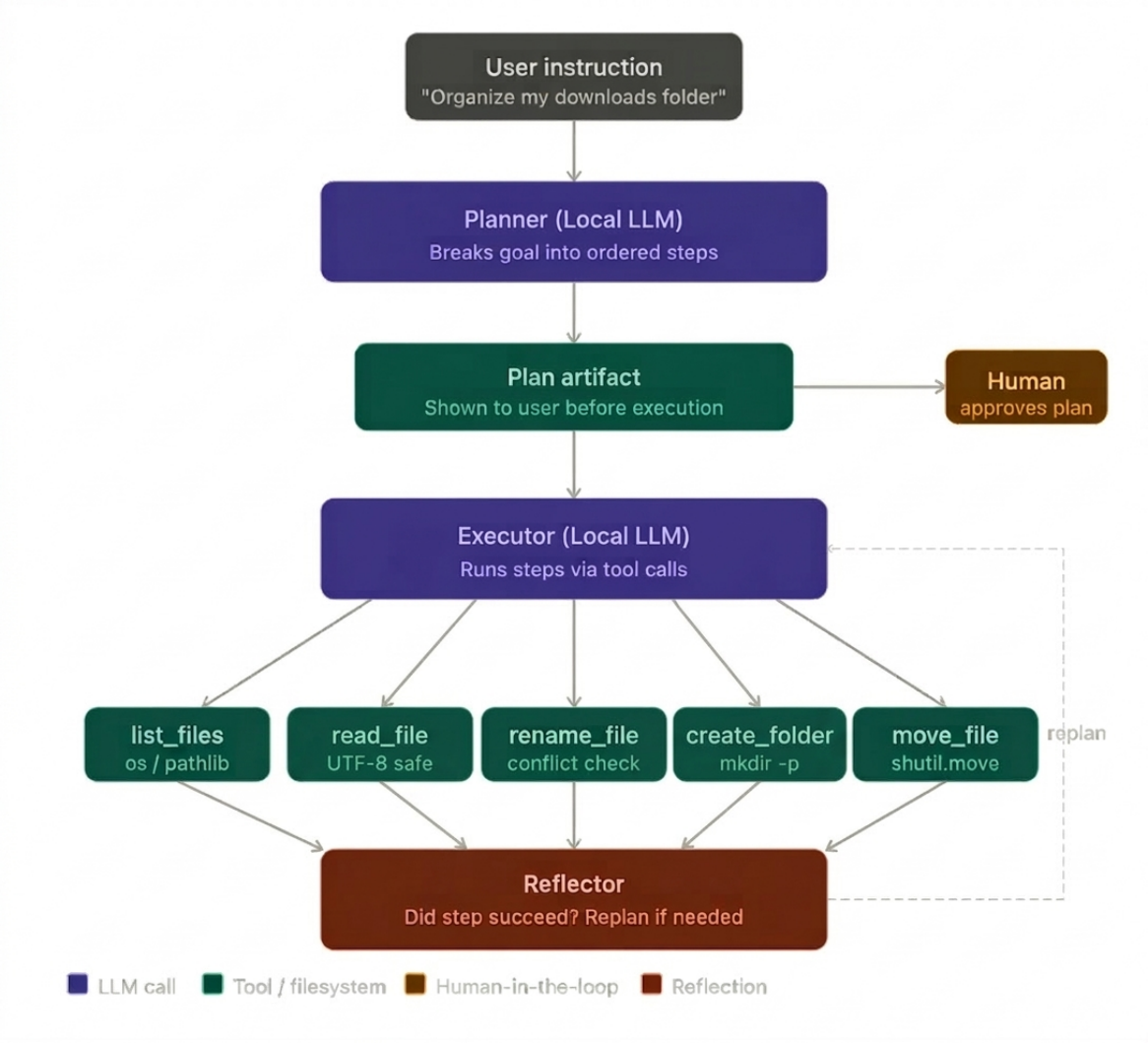

Figure 1. Agent Planning and Execution Flowchart (Generated using Nonbanana)